Lesson 01 Demo 01

Creating a Lambda Function to Launch a Virtual Machine

Objective: To create an AWS Lambda function that automatically launches a virtual machine (EC2 instance) by configuring IAM roles and invoking the EC2 service

Tools required: AWS Management Console

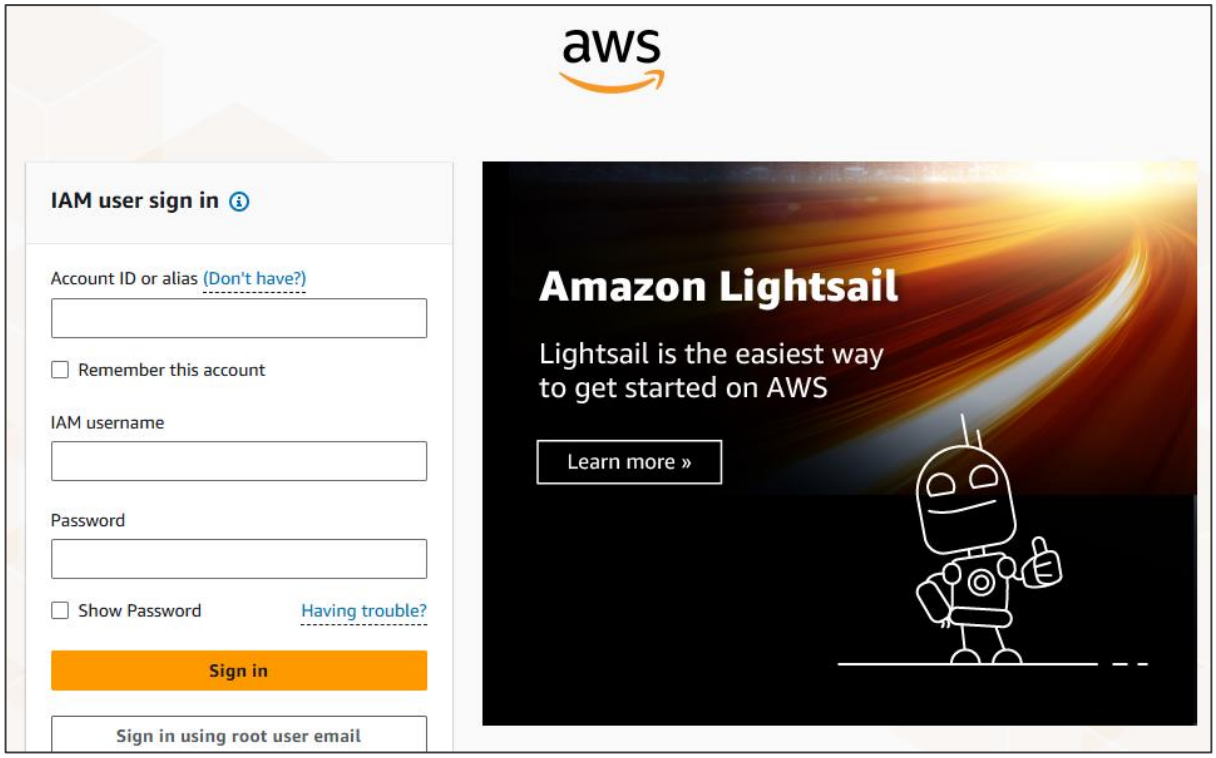
Prerequisites: None

Steps to be followed:

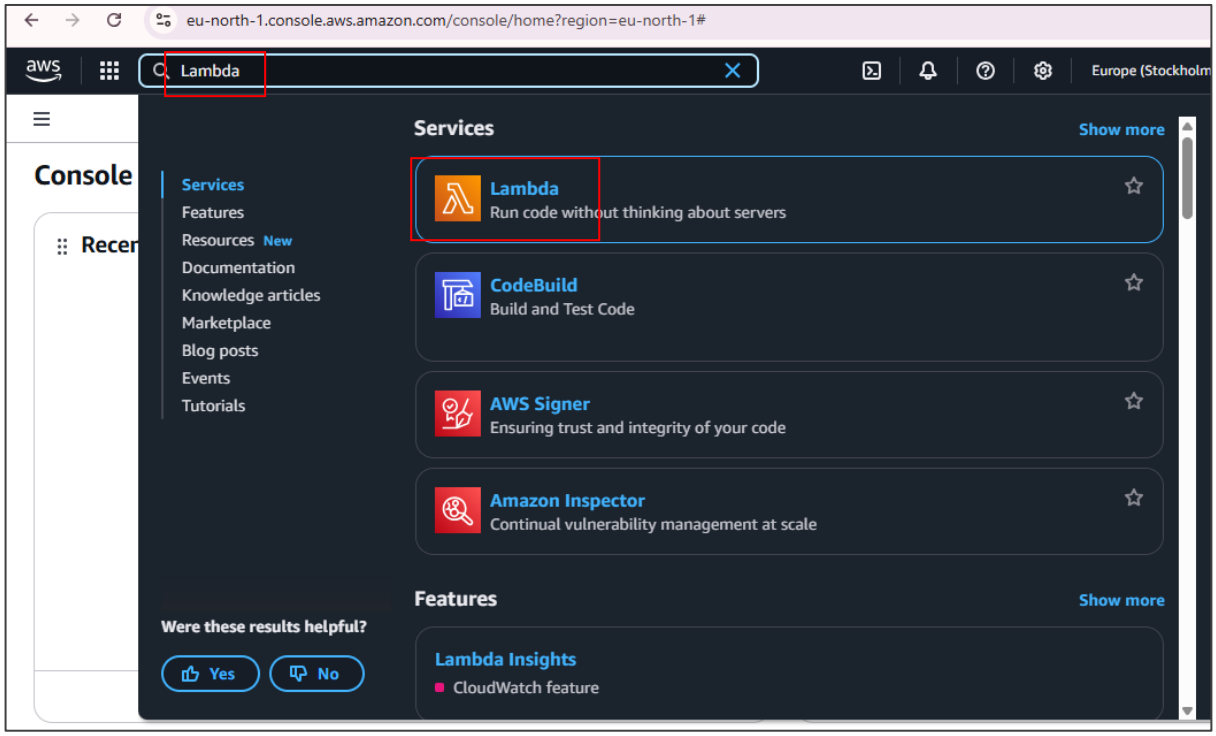
1. Create the Lambda function to launch a virtual machine (VM)
2. Reconfigure IAM role to add EC2 permissions
3. Execute and test the Lambda function

Step 1: Create the Lambda function to launch a virtual machine (VM)

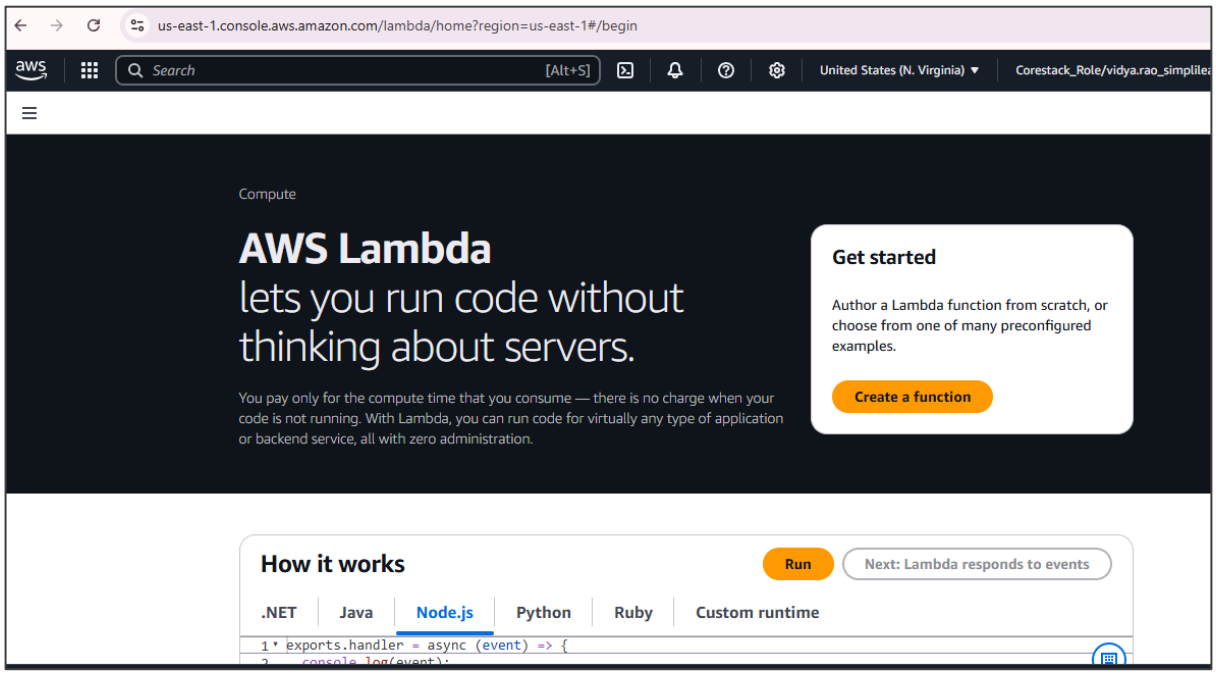
1.1 Sign in to the AWS Management Console



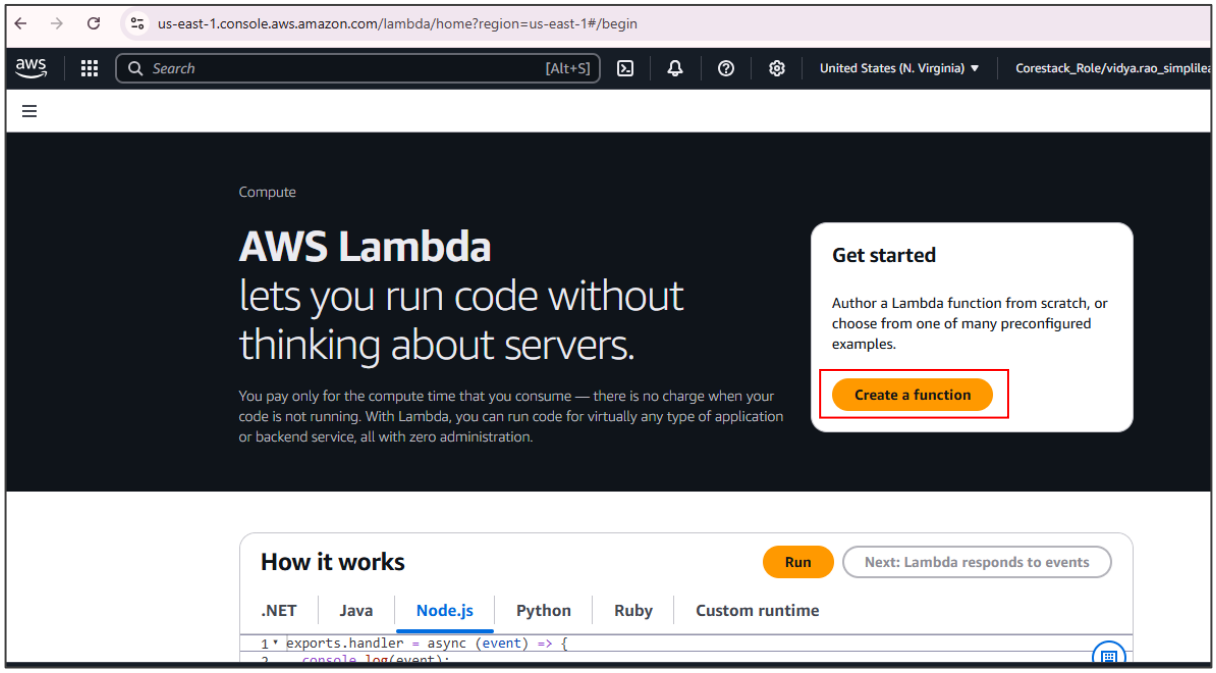
1.2 In the search bar, type **Lambda**, then select the **Lambda** service from the results.



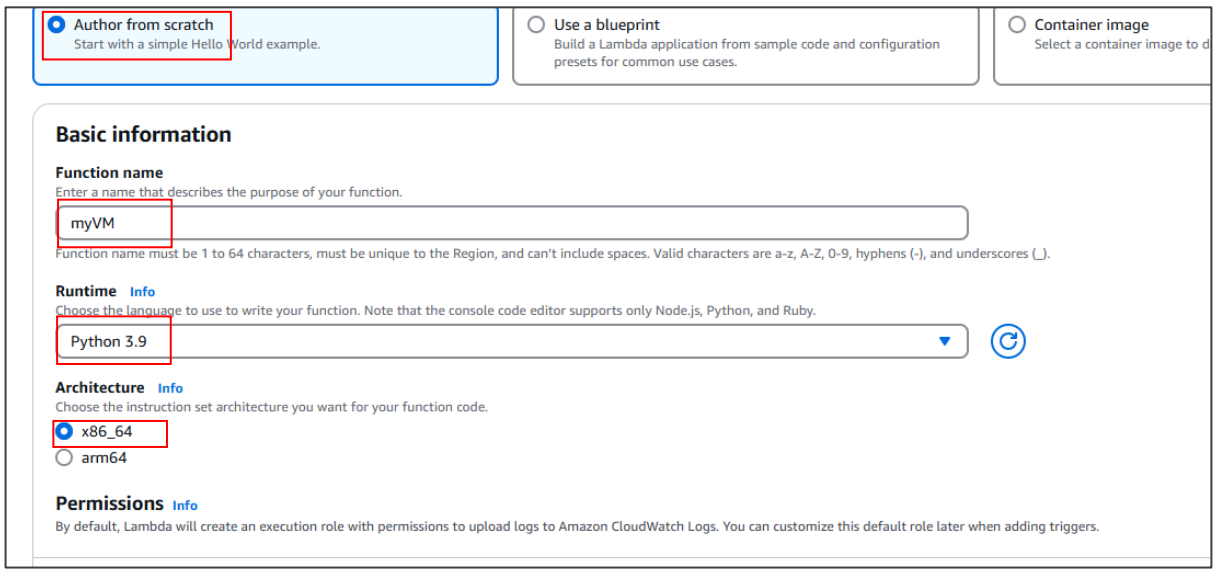
The Lambda service dashboard opens as shown below:



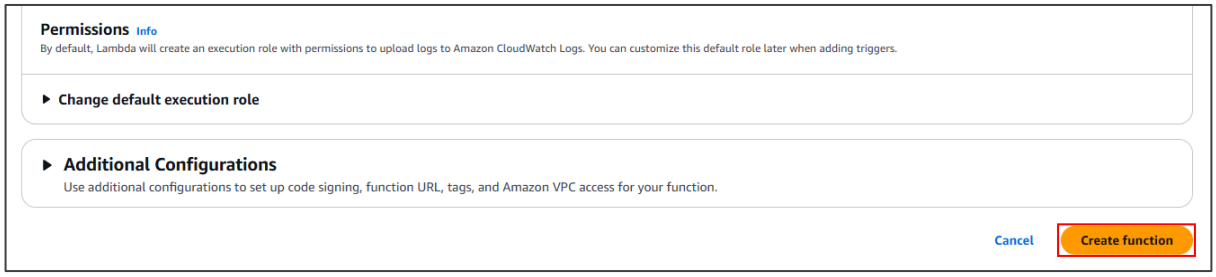
1.3 Click **Create a function**



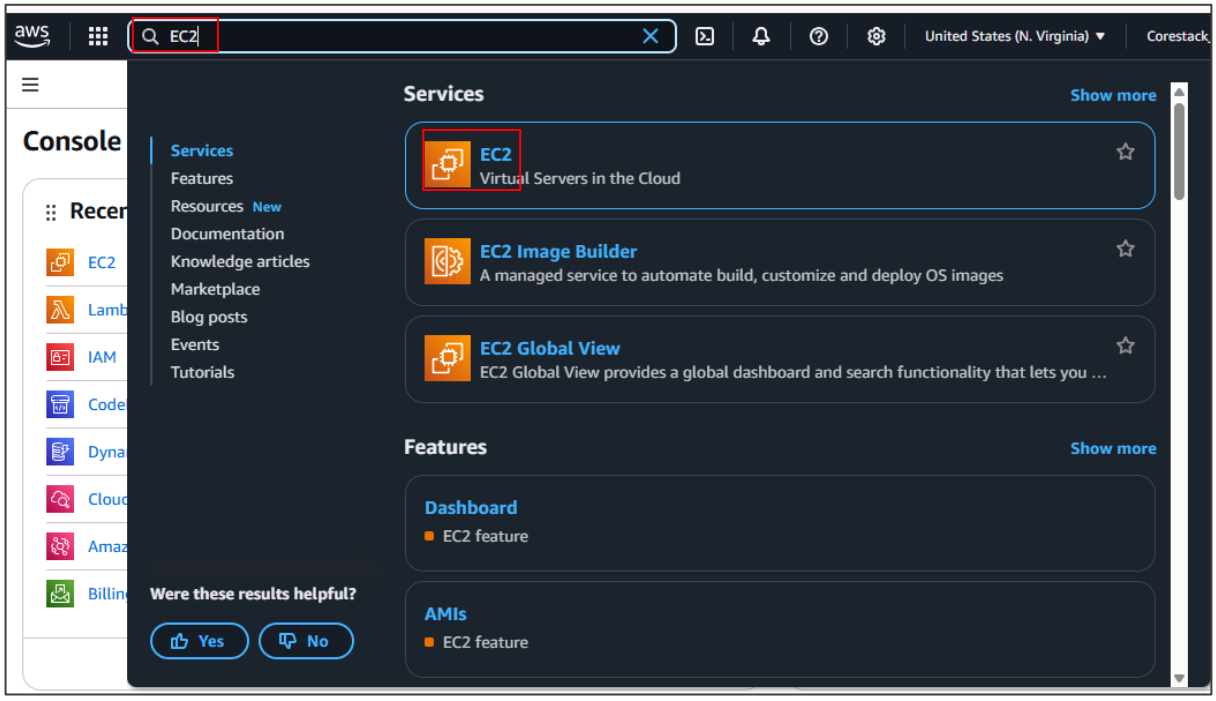
1.4 Click **Author from scratch**, enter the Function name as **myVM**, select **Python 3.9** as the Runtime, and choose **x86_64** as the Architecture



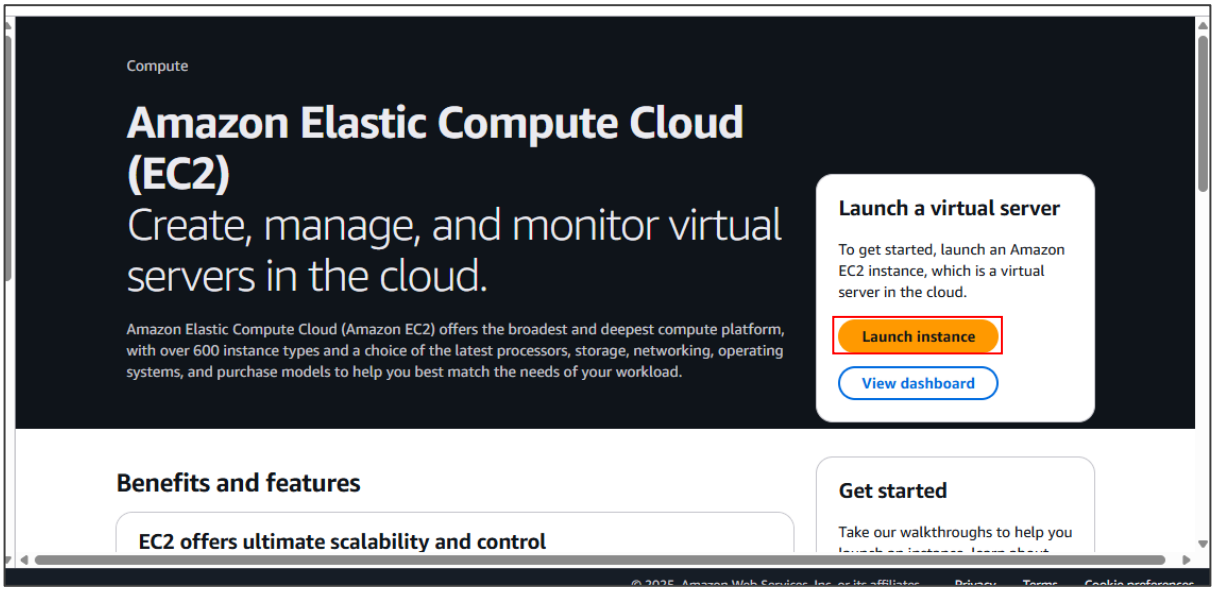
1.5 Click **Create function**



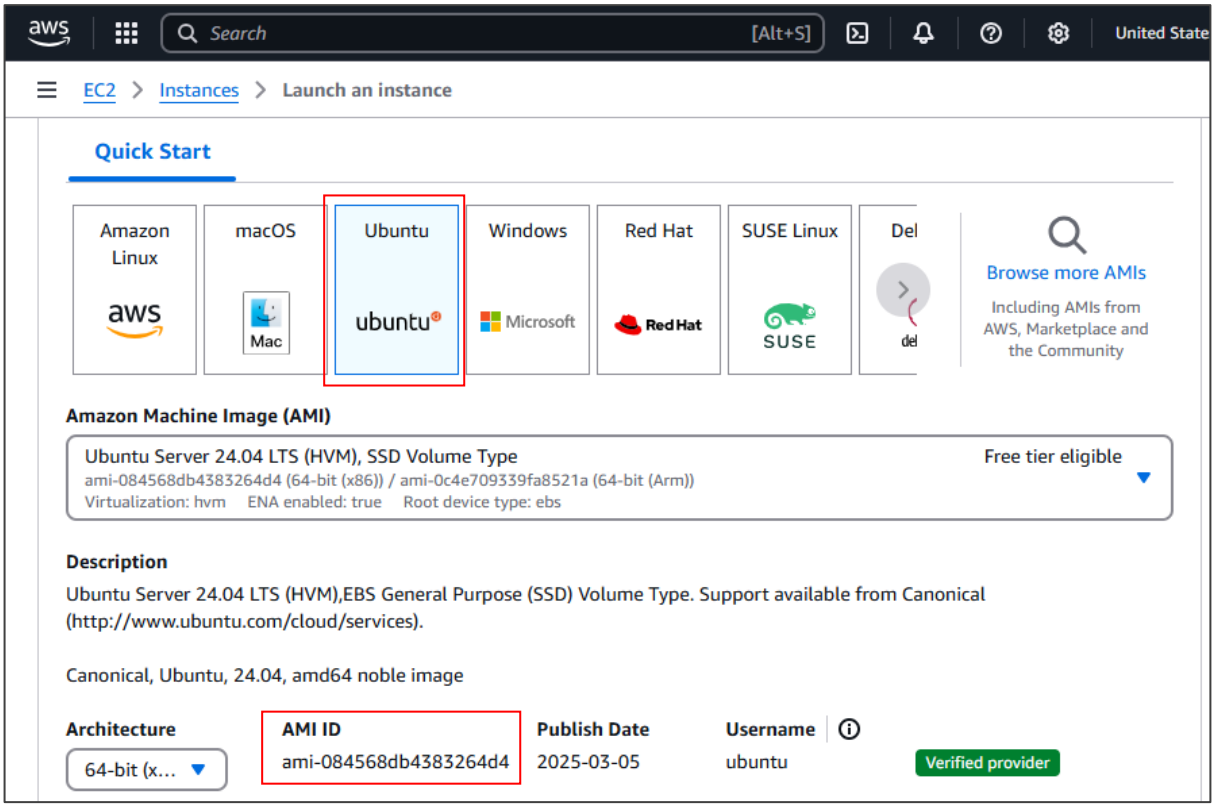
1.6 In the console search bar, type **EC2,** and select the **EC2** service.



1.7 Click **Launch instance** on the EC2 dashboard



1.8 Select an OS type and copy the displayed **AMI ID**



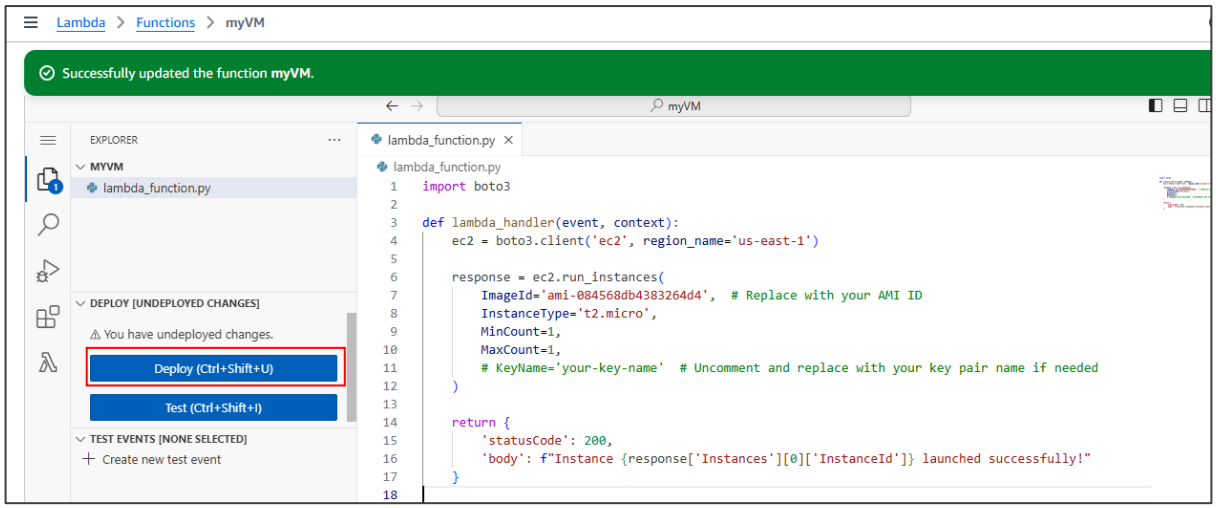
1.9 Navigate back to Lambda service and paste the following code in the code section and click **Deploy**:

```
import boto3

def lambda_handler(event, context):
    ec2 = boto3.client('ec2', region_name='us-east-1')

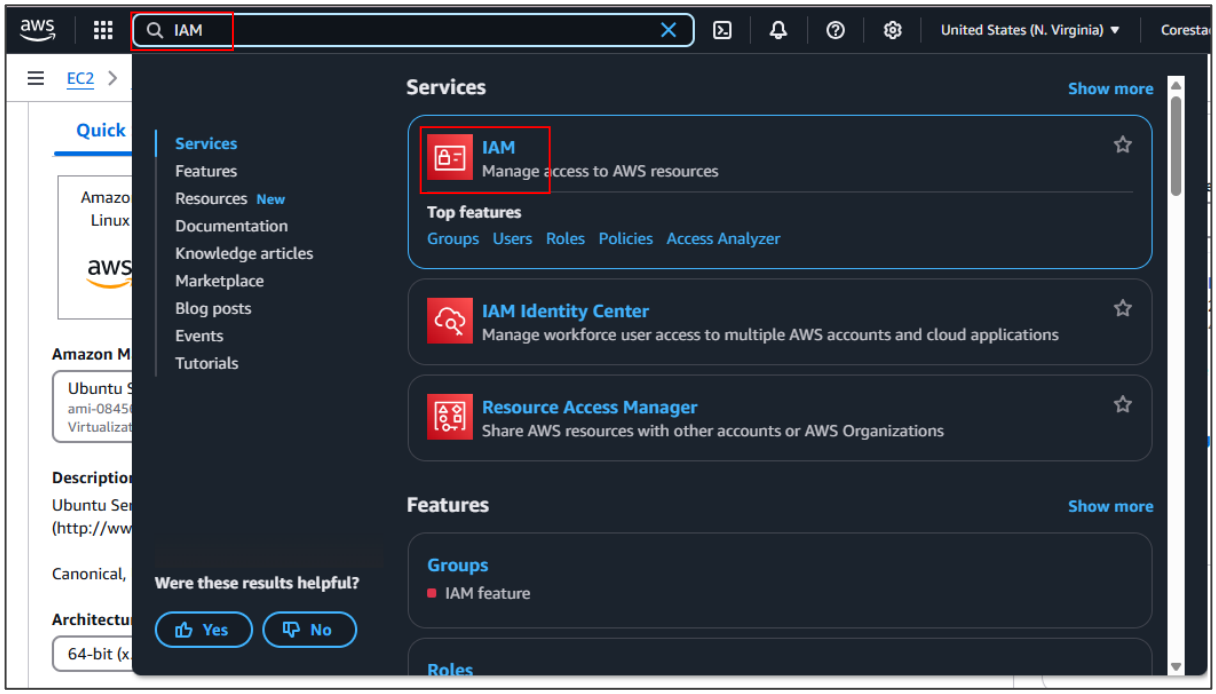
    response = ec2.run_instances(
        ImageId='ami-084568db4383264d4',  # Replace with your AMI ID
        InstanceType='t2.micro',
        MinCount=1,
        MaxCount=1
    )

    return {
        'statusCode': 200,
        'body': f"Instance {response['Instances'][0]['InstanceId']} launched successfully!"
    }
```

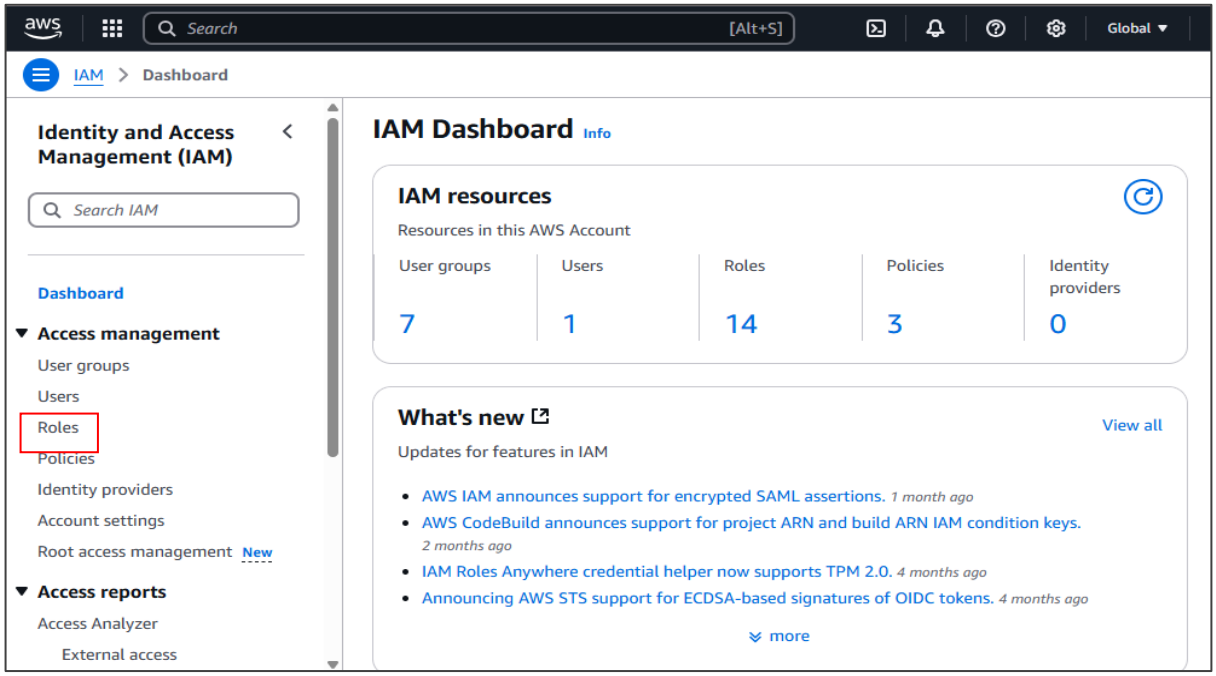


Step 2: Reconfigure IAM role to add EC2 permissions

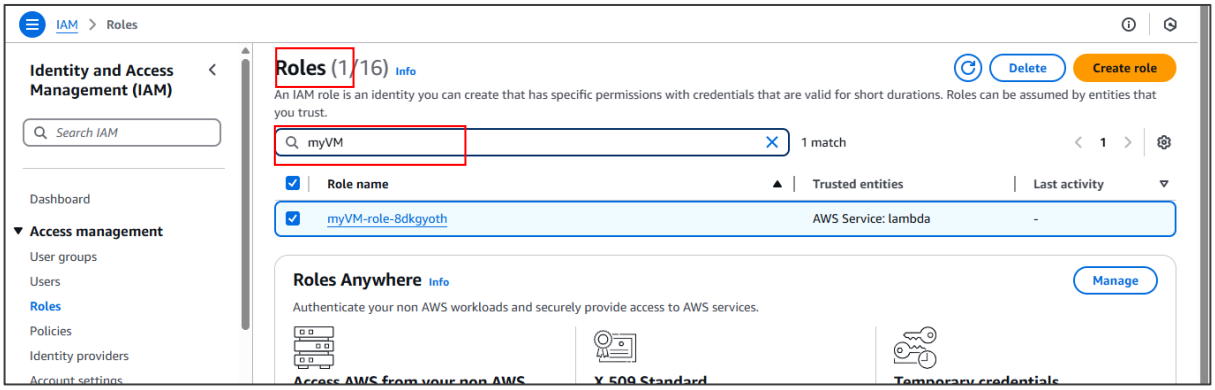
2.1 Search for **IAM** in the search bar, and select the **IAM** service from the list of results



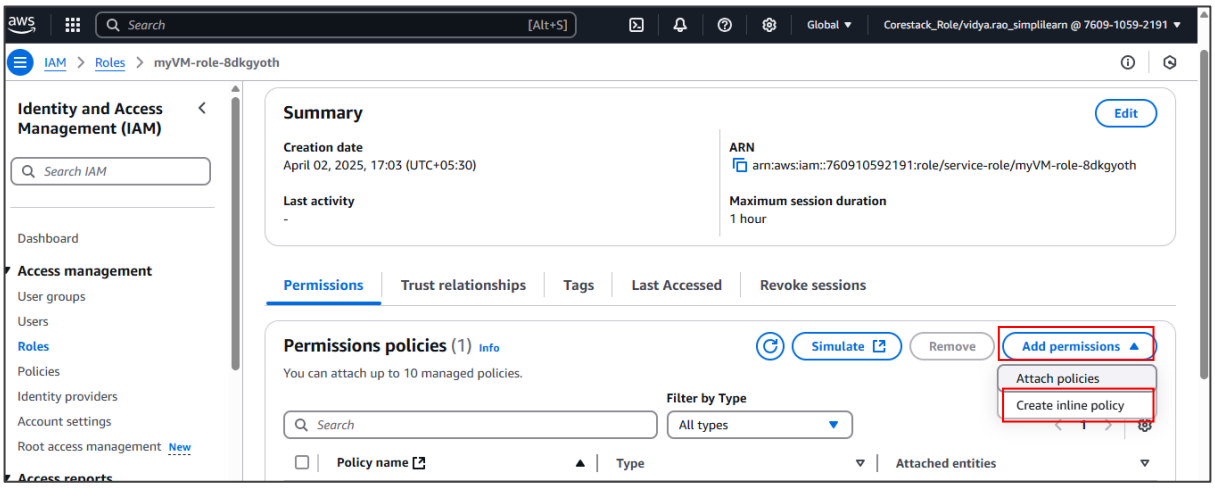
2.2 Click **Roles** under the Access management section



2.3 Locate your Lambda execution role by name



2.4 Click **Add permissions**, then select **Create inline policy** to define custom EC2 permissions for the Lambda role

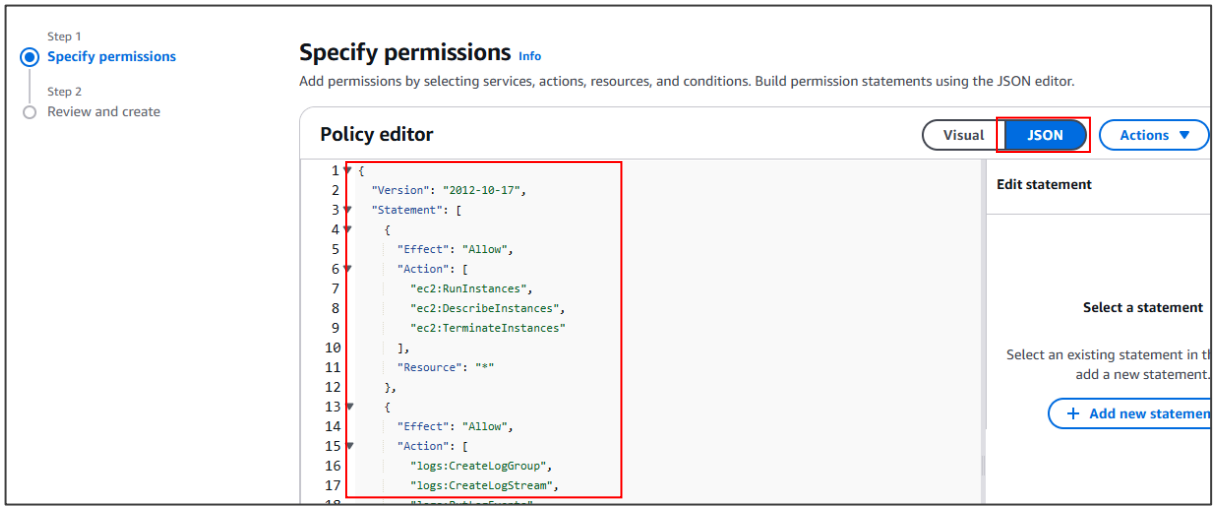


2.5 On the create policy page, click the **JSON** tab, add the following script, and click next:

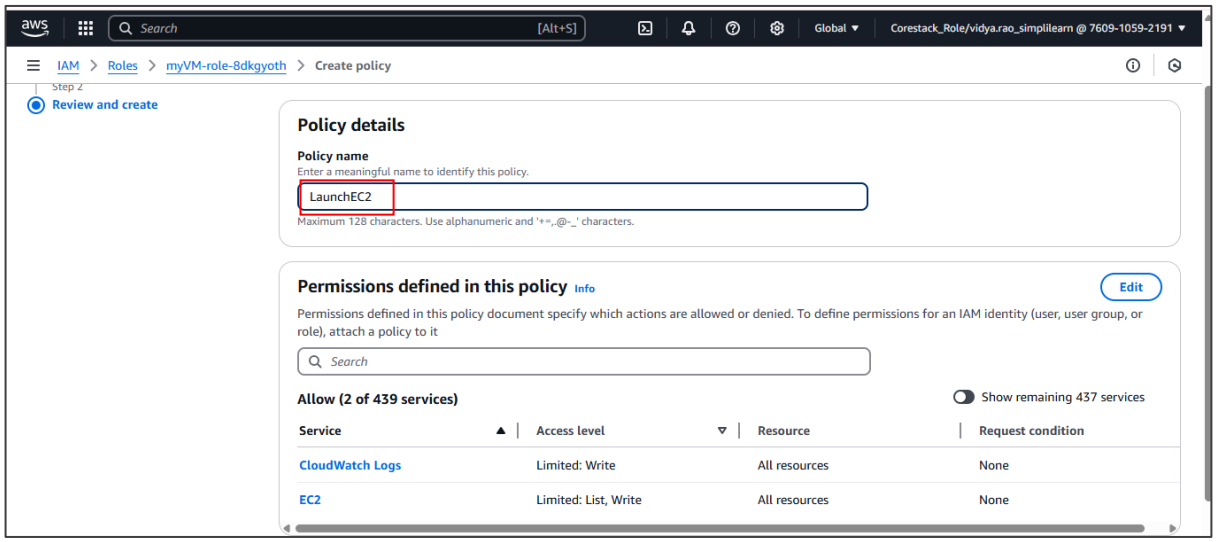
```
{

 "Version": "2012-10-17",
 "Statement": [
  {
   "Effect": "Allow",
   "Action": [
    "ec2:RunInstances",
    "ec2:DescribeInstances",
    "ec2:TerminateInstances"
   ],
   "Resource": "*"
```

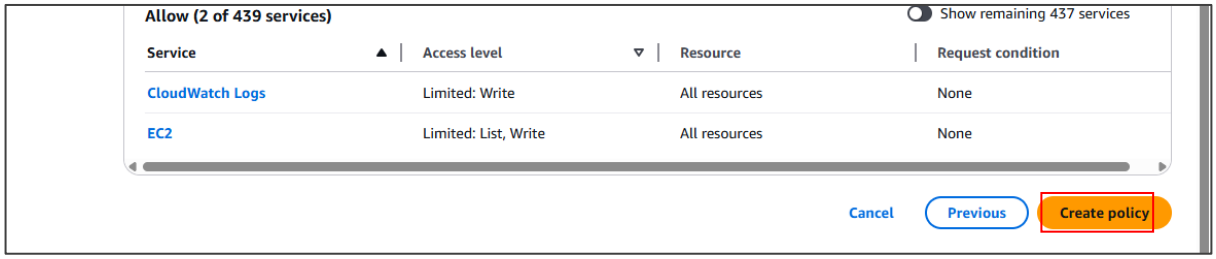
```
    },
    {
      "Effect": "Allow",
      "Action": [
        "logs:CreateLogGroup",
        "logs:CreateLogStream",
        "logs:PutLogEvents"
      ],
      "Resource": "*"
    }
  ]
}
```



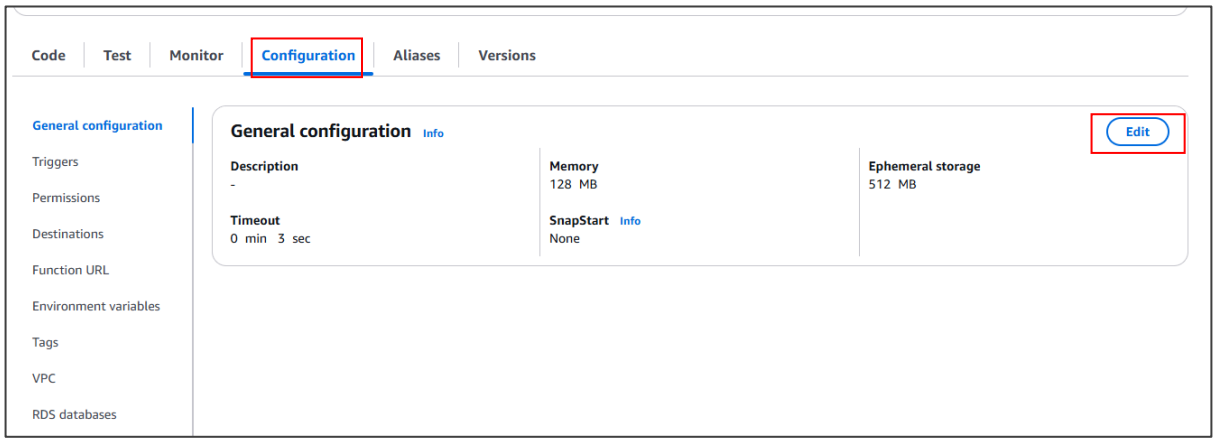
2.6 Enter the Policy name as **LaunchEC2**



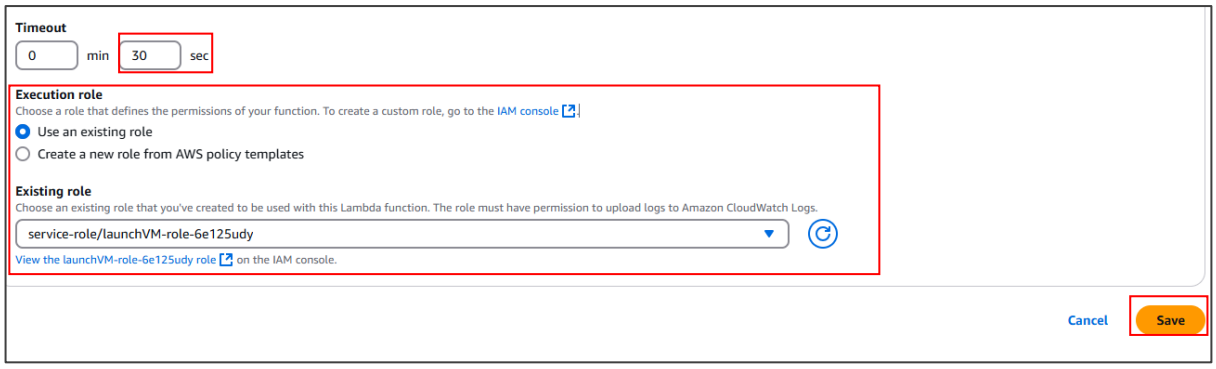
2.7 Click **Create policy**



2.8 Return to the Lambda service and click **Edit** in the **Configuration** section

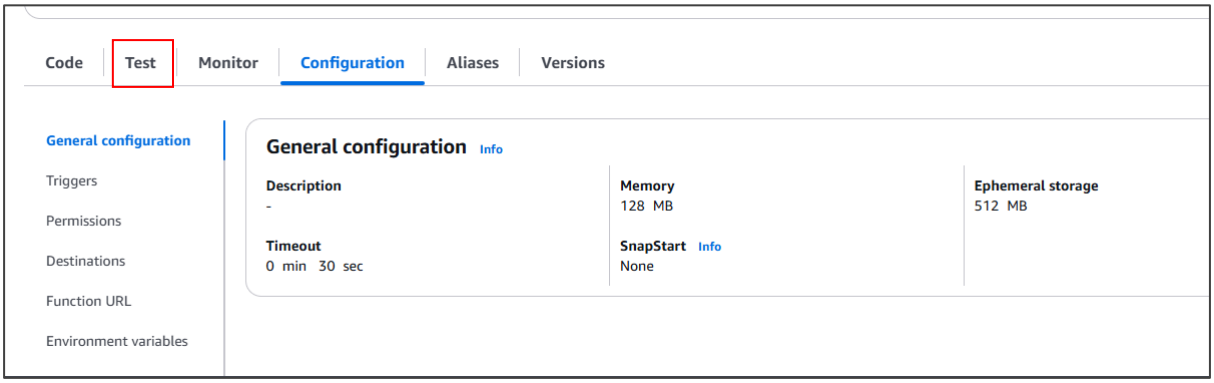


2.9 Set the timeout duration to 30 seconds, select the existing execution role, and click **Save** to apply the configuration

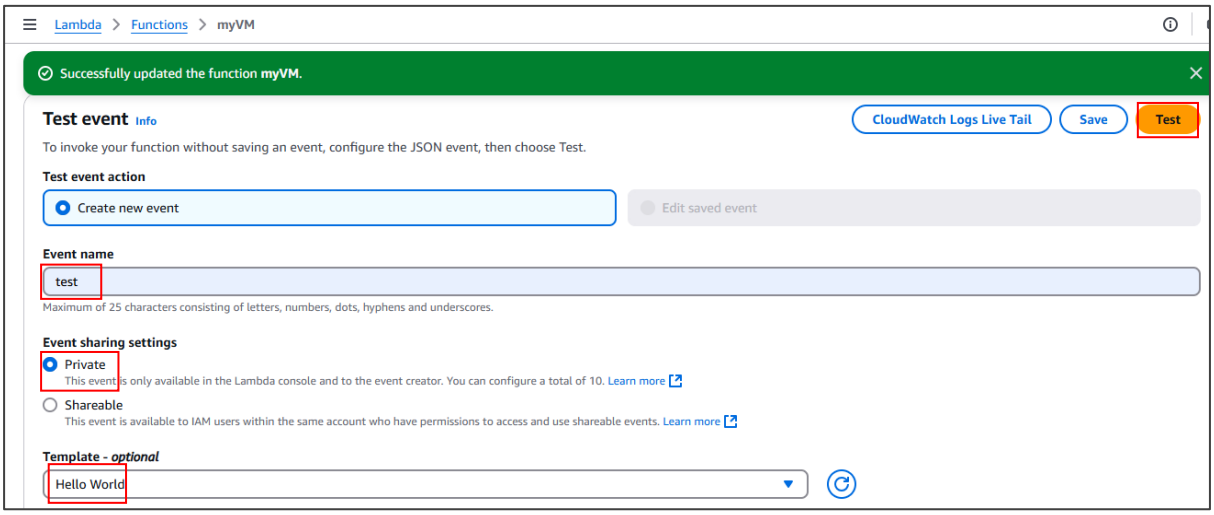


Step 3: Execute and test the Lambda function

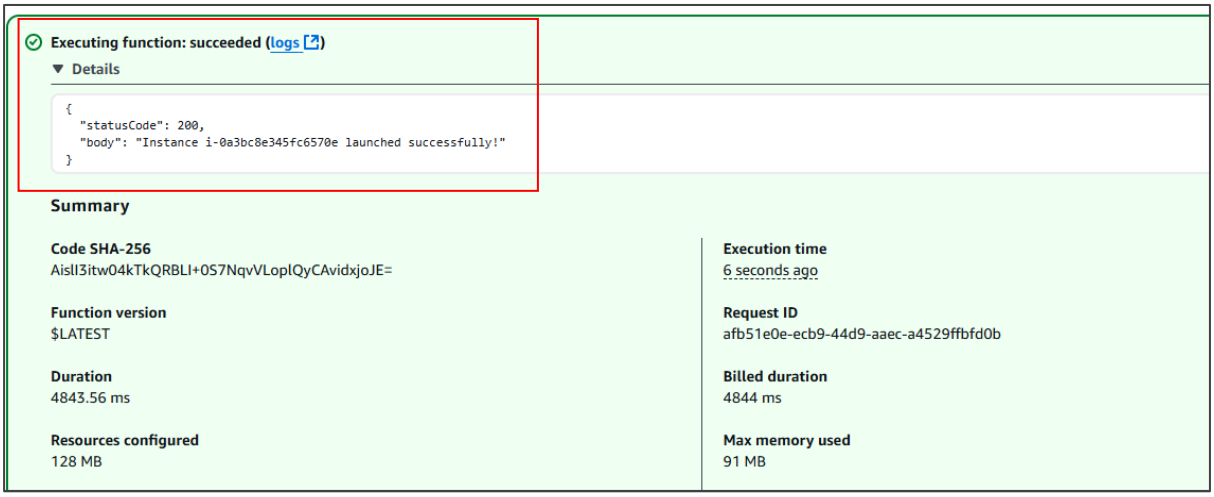
3.1 Click **Test** to run the Lambda function



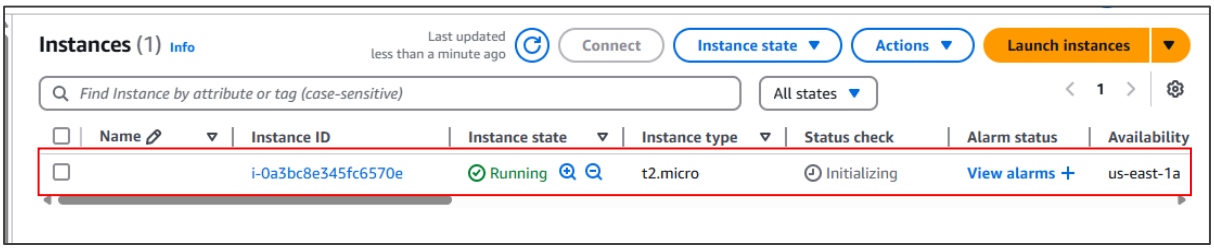
3.2 Enter a name for the **test** event, keep the event **Private**, choose the **Hello World** template, and click **Test** to execute the function



3.3 Verify the execution result to confirm a status code of 200



3.4 Check that the instance has been successfully launched



By following these steps, you have successfully created and executed an AWS Lambda function that launches an EC2 instance. This process demonstrated how to integrate Lambda with EC2 and manage role-based access using IAM.